NONLINEAR PRICING

Pricing Strategies
2024

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Overview

Customer likely do not value the next incremental unit same as the previous unit.
Law of diminishing return.
- So; we can price discrim between each unit that a same customer buy.
- and how to price discrim between each customer base on quantity.

contrast; with yesterday, discrim base on quality (product line feature)

Definition: ***Nonlinear Pricing*** *refers to pricing structures with a nonlinear relationship between the price charged and the quantity of goods/services sold.*

- Price discrimination between each unit sold to a customer
- Managing a menu of non-linear pricing schedules boils down to **product line pricing** for segmentation

➡ Use quantity as "*fence*" (instead of quality) to capture value

NL Pricing Examples

First 3; cosmetically looks very different, but economics wise, they are very similar.

subscription; also a 2 part tariff where the variable fee is set to zero. (slightly different)

- Credit Card Pricing
  - Monthly Fee & Interest Rate

2-part tariff

average price i pay (per unit of credit consumed) reduces as i spend more

- Perdue Frozen Chicken
  - Regular Pack: $4.99/lb.
  - 5 pound bag: $3.99/lb

Volume discount
("Commodity Bundle")

- Loyalty Program
  - lower price on consecutive units

Quantity discount schedule
("Incremental units pricing")

- Amazon's *Kindle Unlimited* program
  - $9.99/month
  - Unlimited access to over 3 million eBooks

subscription

NL Pricing as Segmentation

- Look for variation in WTP for each incremental unit demanded
- Two economic rationales for using non-linear pricing to segment:

1. **Targeted Pricing** with a single customer
2. **Product Line Pricing** with different customer segments

See Dolan (1987) for a more formal treatment of nonlinear pricing

Strategies to *capture* value

Two Illustrative Examples

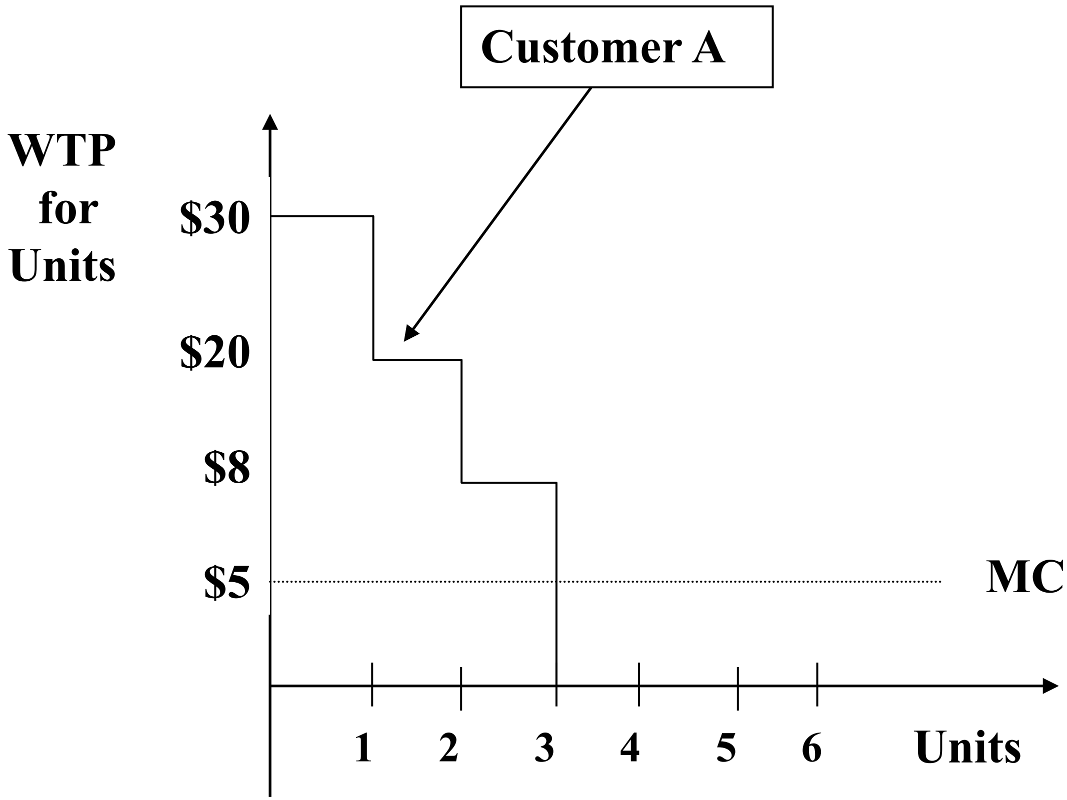
1 **Single Customer**

- Illustrate 3 ways of capturing surplus

2 **Two Customers:** High Volume/Value vs. Low Volume/Value

- Illustrate threat of "trading down"

Example 1:
Single Customer, potentially buys up to 3 units



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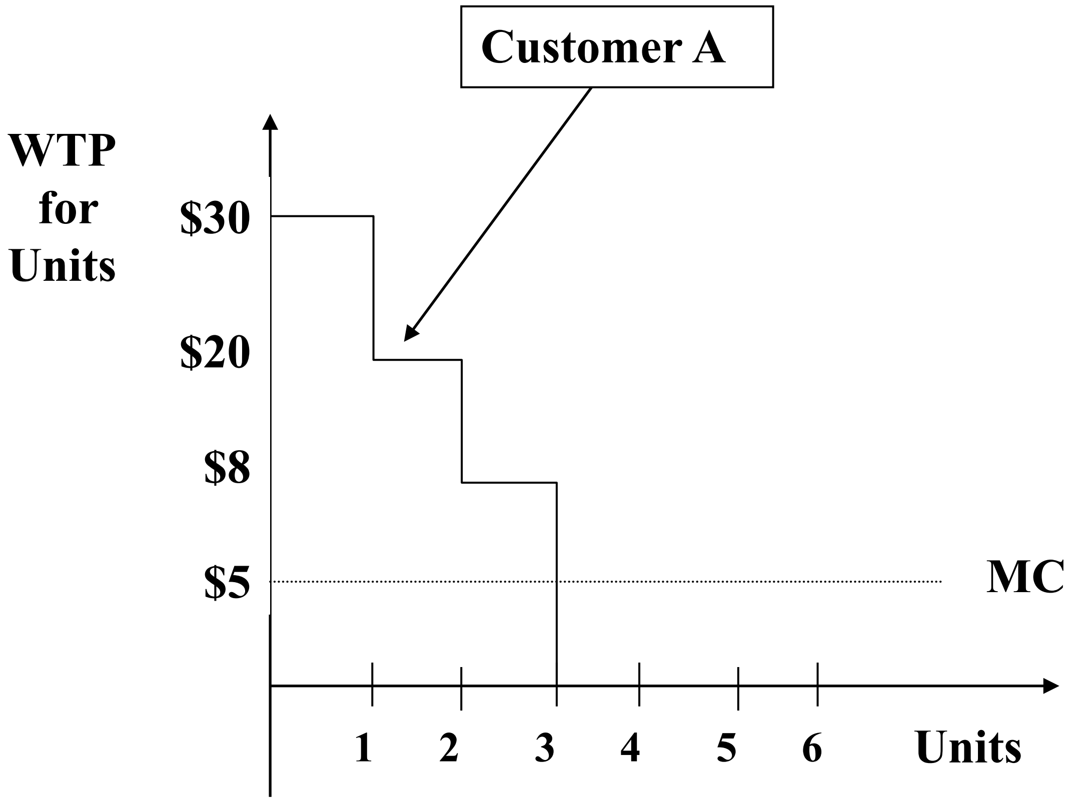

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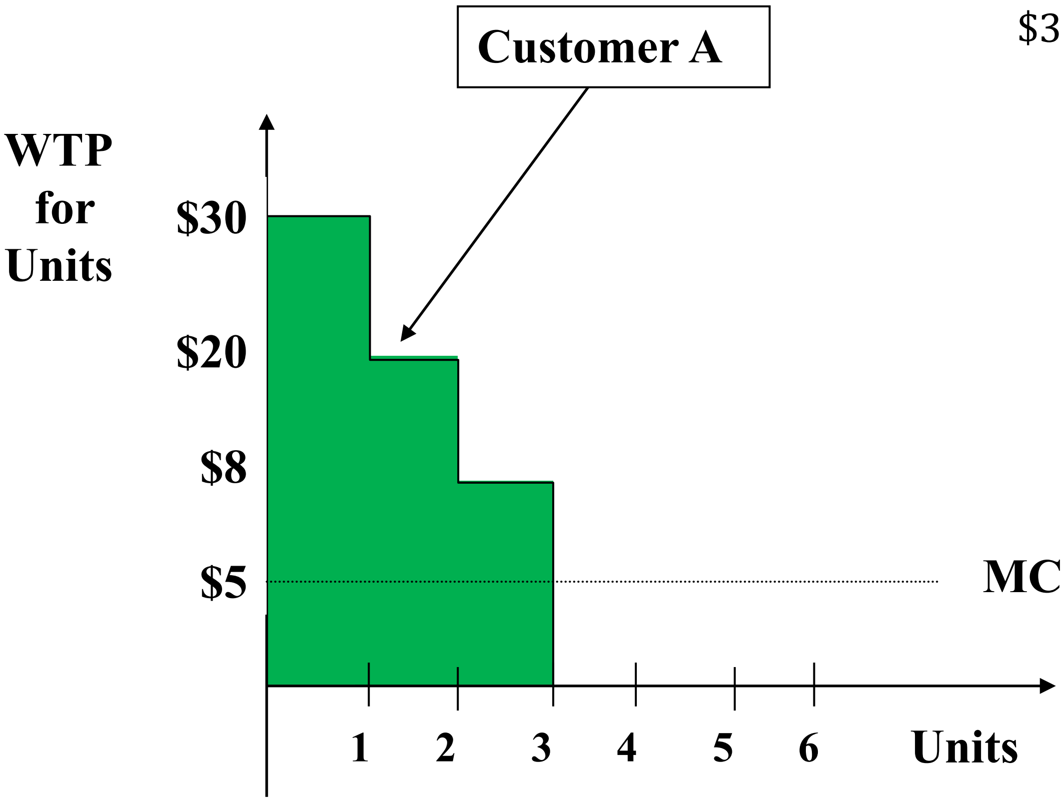

Total Customer A Value:

Total WTP for Customer A = Total Surplus

$\$30 \times 1\ +\ \$20 \times 1\ +\ \$8 \times 1\ =\ \$58$

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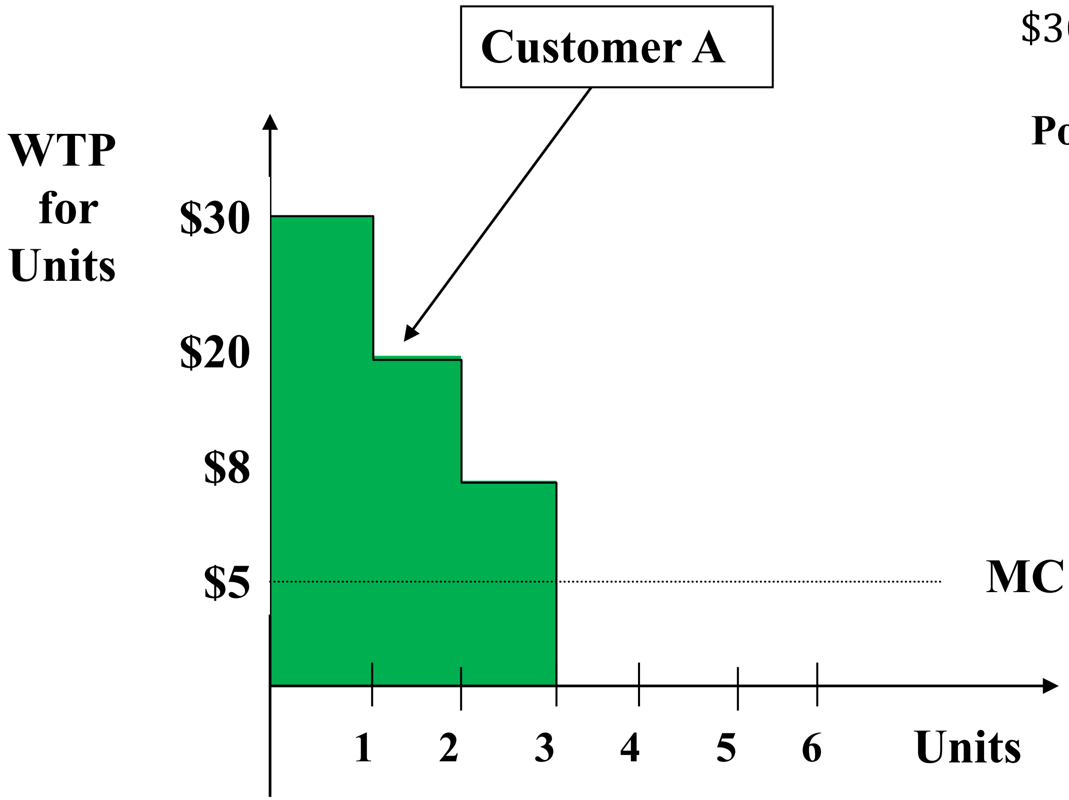

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Potential Profit from Customer A:

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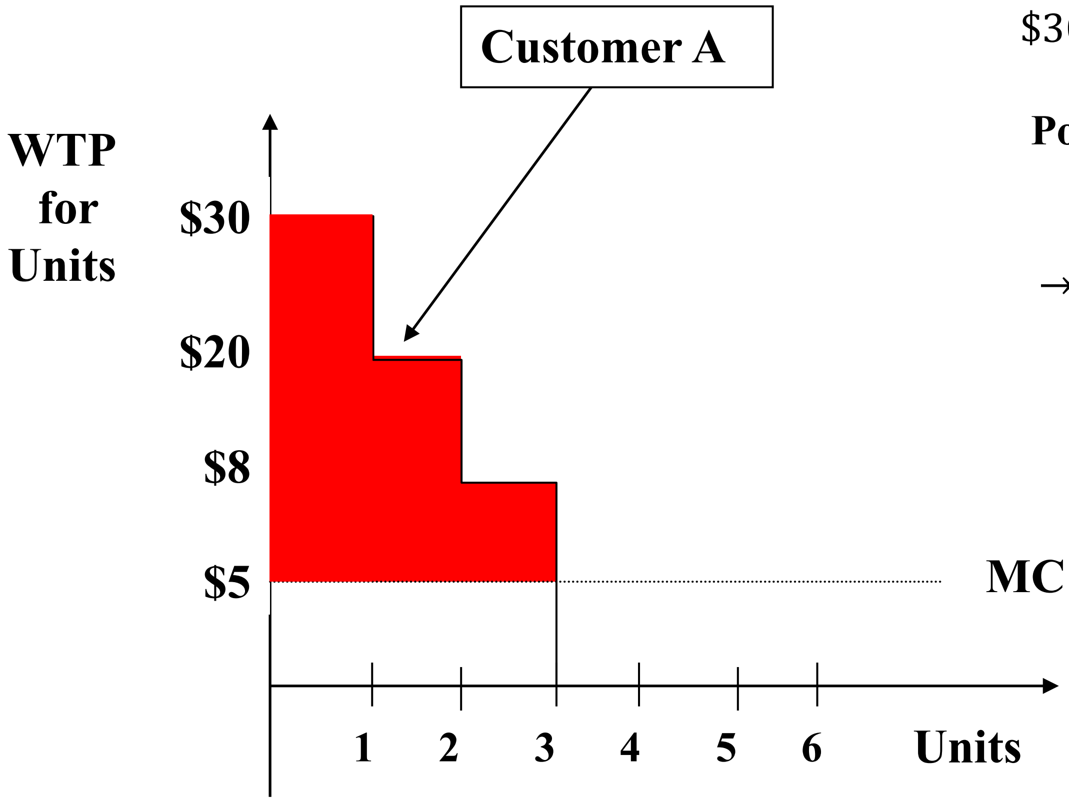

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Potential Profit from Customer A:

Cost to Produce 3 units

$\$5 \times 3 \ = \$15$

$\rightarrow \$58 - \$15 \ = \ \$43$

Demand curve (of customer A) is the EVC that he derives.
Anything above zero line is his WTP / EVC.

Max profit is above MC = 43

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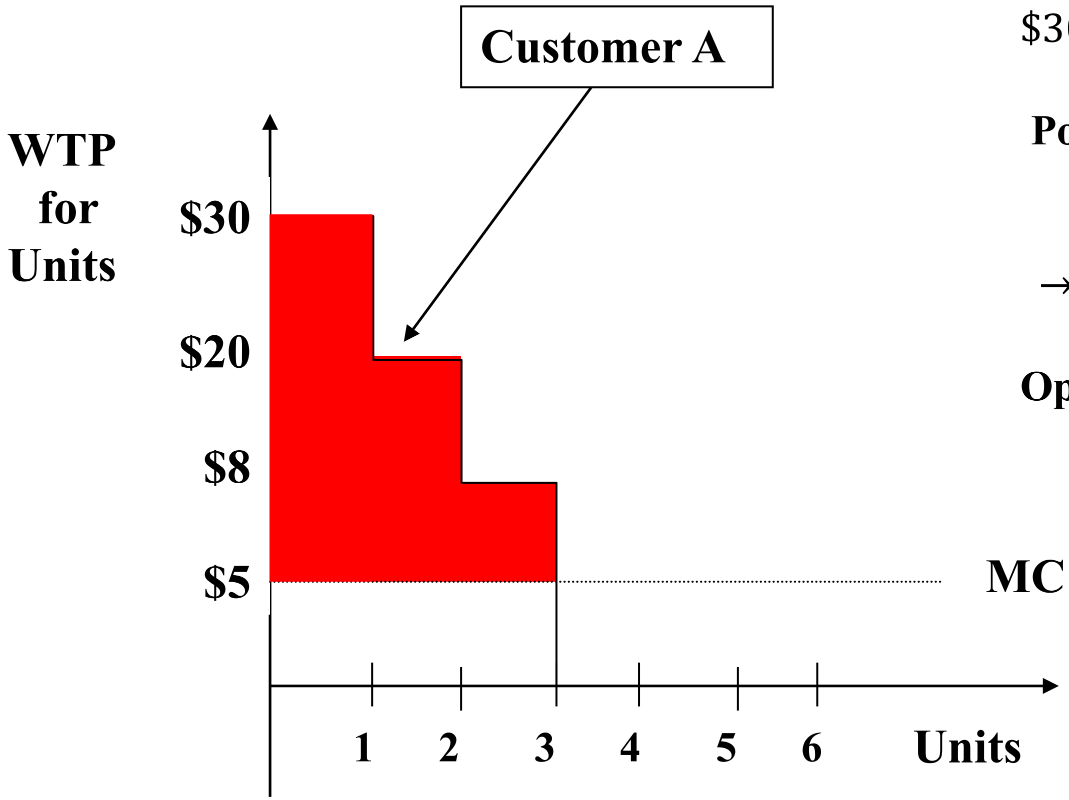

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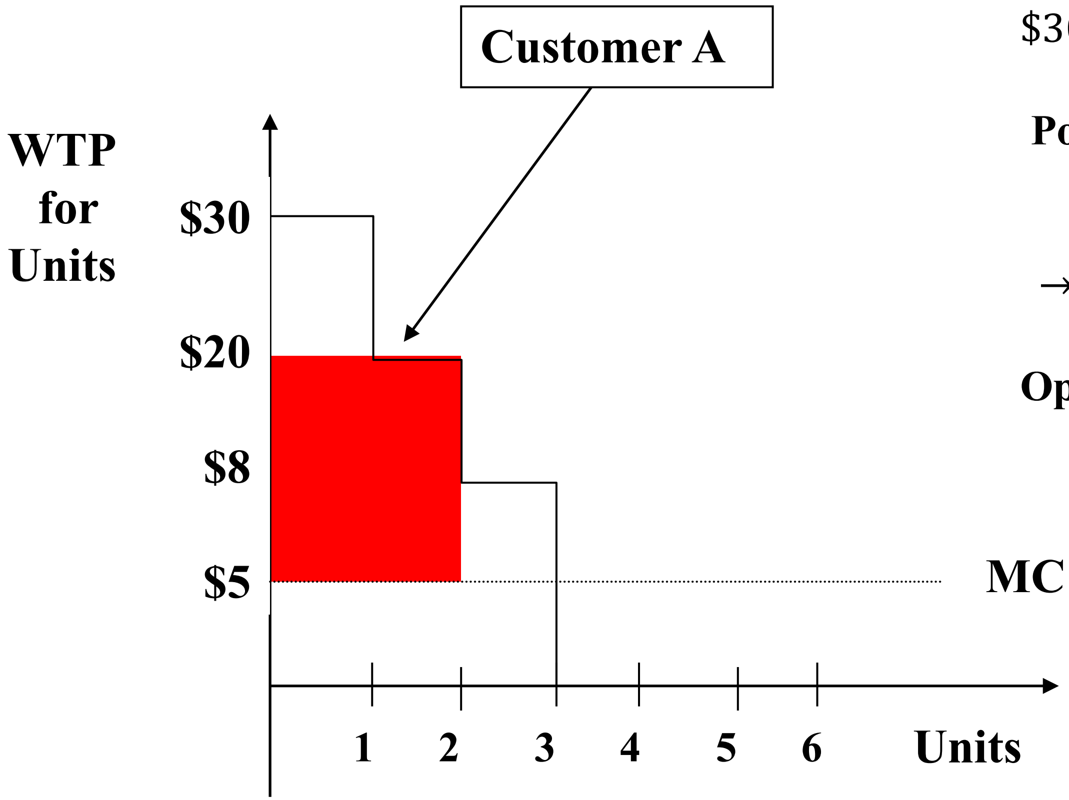

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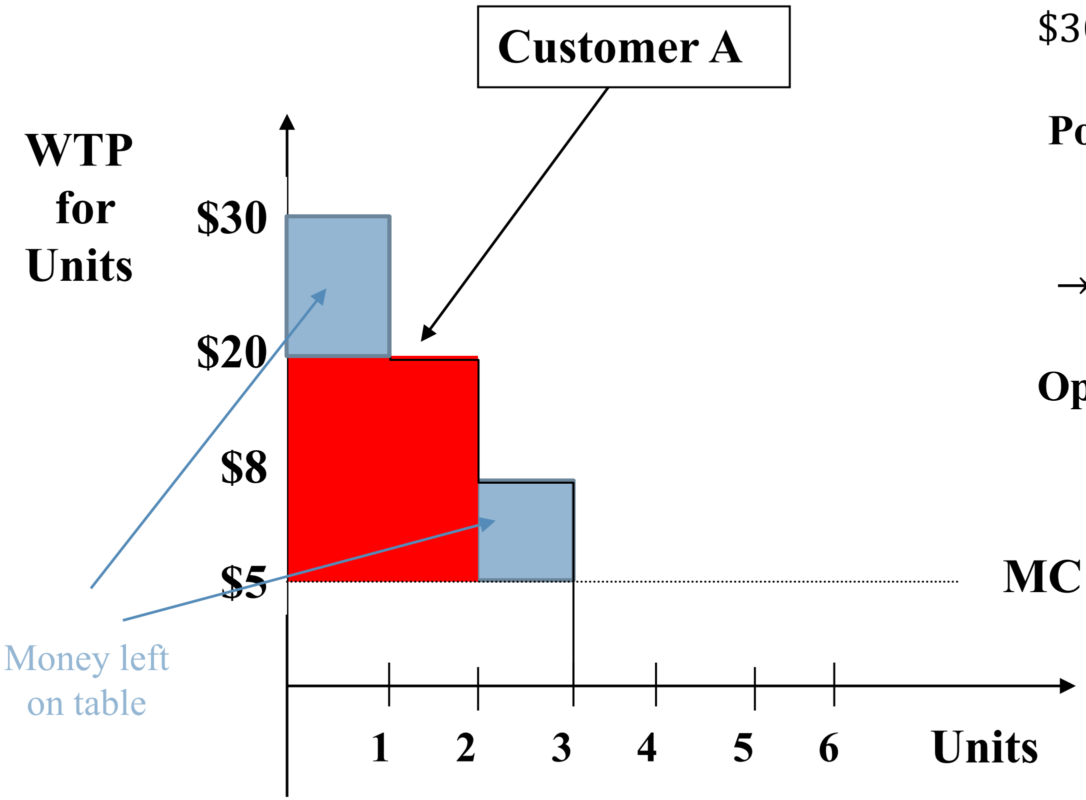

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we are leaving $13 potential profit on the table.

Lets use NL pricing to capture the value.

Example 1:

Two Strategies to capture value

1 *Quantity Discount Schedule (or incremental units discounts)*

- First Unit: $30
- Second Unit: $20
- Third Unit: $8

2 *Commodity Bundle* (*i.e.*, Take-it-or-leave-it)

- 3 units, $58

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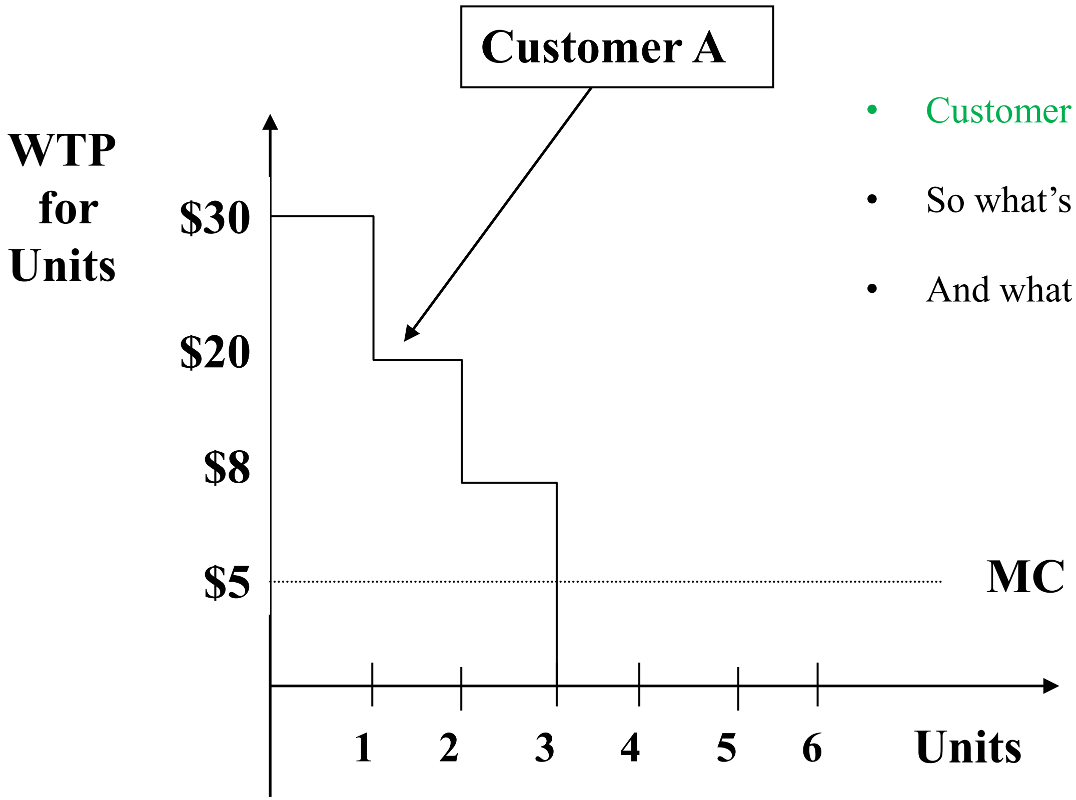
- What about a **Two-Part Tariff**?

$$\underbrace{T(Q)}_{\text{Total charged}} = \underbrace{F}_{\text{Fixed fee}} + \underbrace{pQ}_{\text{Variable fee}}$$

Example 1:
Two-Part Tariff

$$T(Q) = F + pQ$$



- Customer's surplus (CS) determines Fee, ***F***, you can charge
- So what's best *feasible* variable price, p, to maximize CS?
- And what fee does that let you charge?

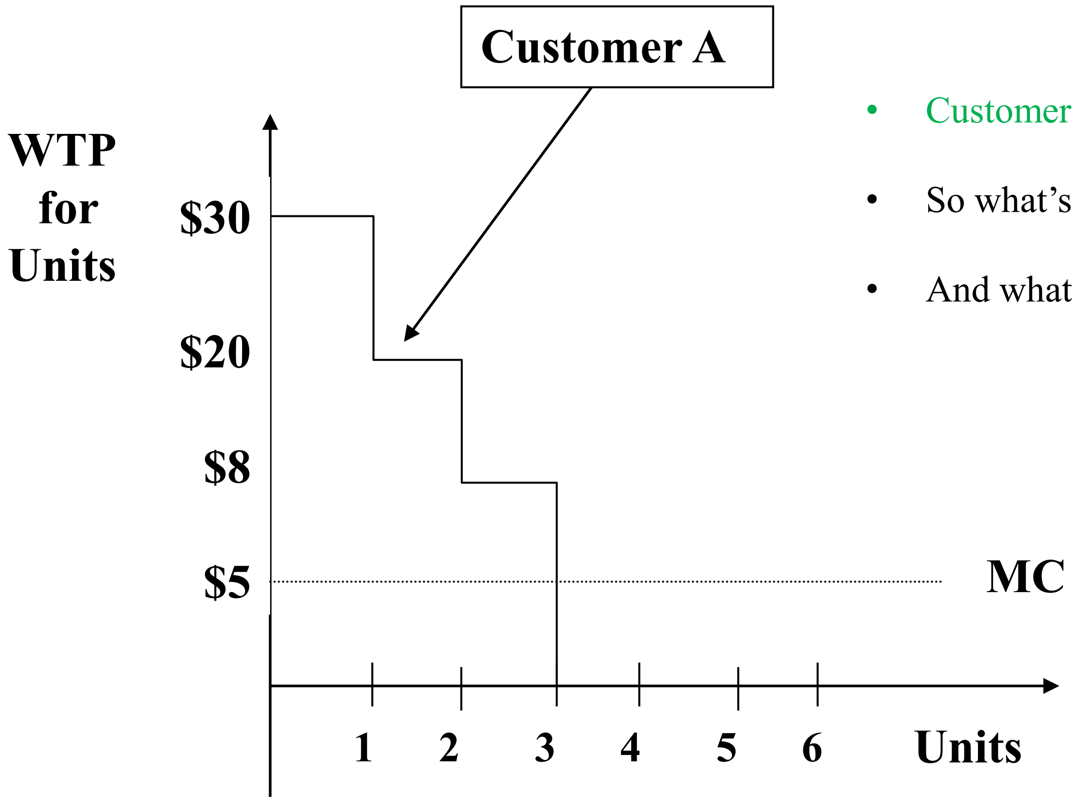
Example 1:
Two-Part Tariff

Charging all of the profit $43 upfront (as a fixed fee), and then charging a small per unit variable price.

Access price

usage price

$$T(Q) = F + pQ$$



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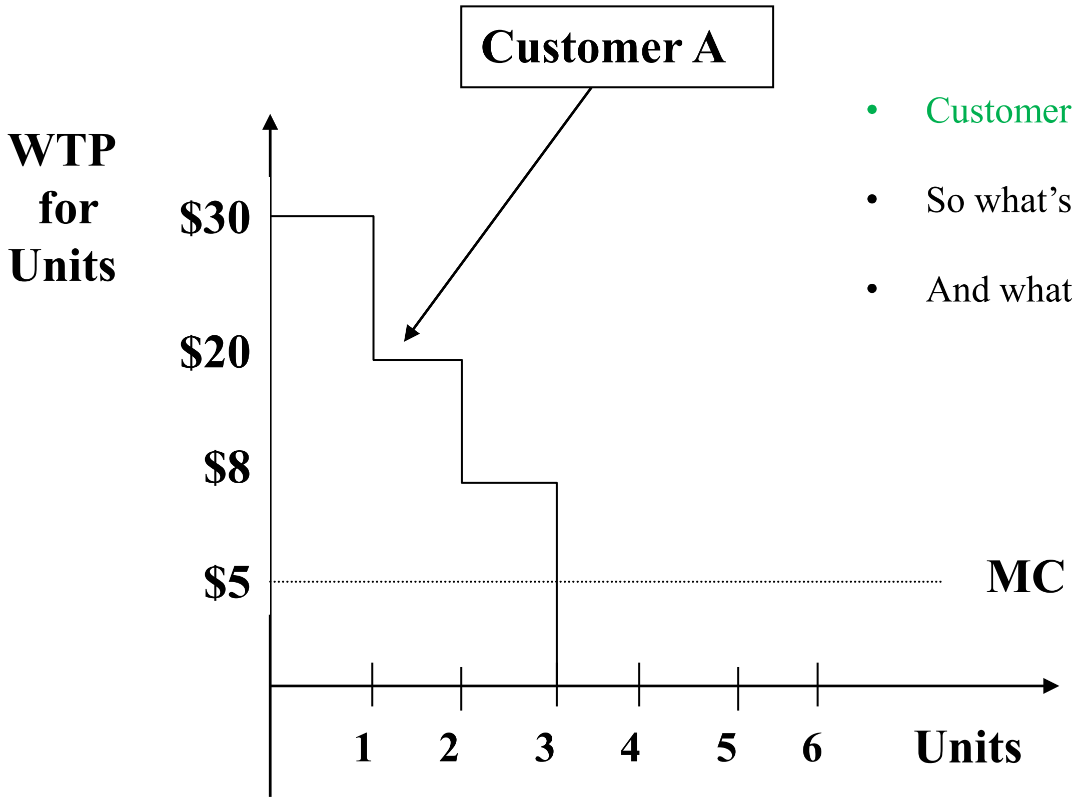
Example 1:
Two-Part Tariff

Access price usage price

$$T(Q) = F + pQ = 43 + 5Q$$



- Customer's surplus (CS) determines Fee, F, you can charge
- So what's best *feasible* variable price, p, to maximize CS?
- And what fee does that let you charge?

Example 1:
Three Strategies to capture value

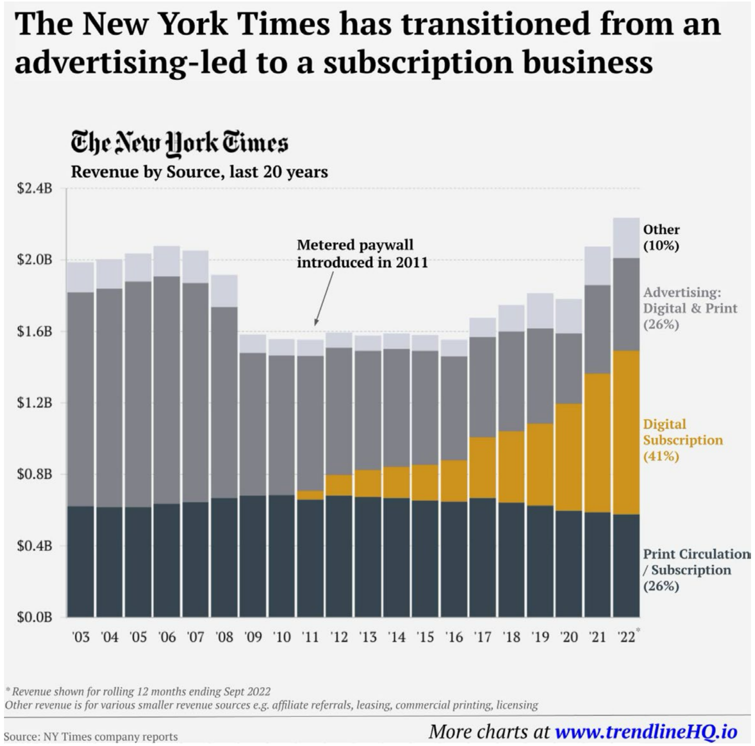
1. *Quantity Discount Schedule*
   - 1st Unit: p_1=\$30
   - 2nd Unit: p_2=\$20
   - 3rd Unit: p_3=\$8

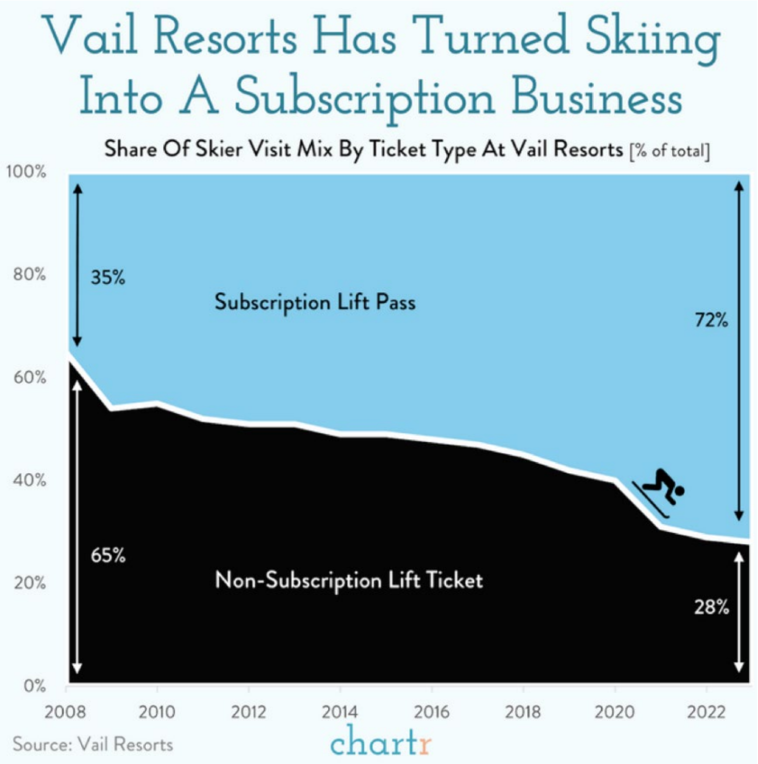
2. *Commodity Bundle* (*i*.e. Take-it-or-leave-it)
   - 3 units, \$58

3. *Two-Part Tariff*
   - p=\$5 per unit (bill at **marginal cost**)
   - F = \$25+\$15+\$3 = \$43 entry fee (profit as up-front down-payment)

Two-Part Tariff with digital goods
Subscription

- If marginal cost = $0, you have a pure subscription model!





Challenges: Arbitrage Risks

- Resale exclusion may be important to avoid competition from customers acting as secondary suppliers in resale markets
  - Bulk-buying and re-selling (grey markets)
  - Forward-buying

Grey market? result from excess buying.

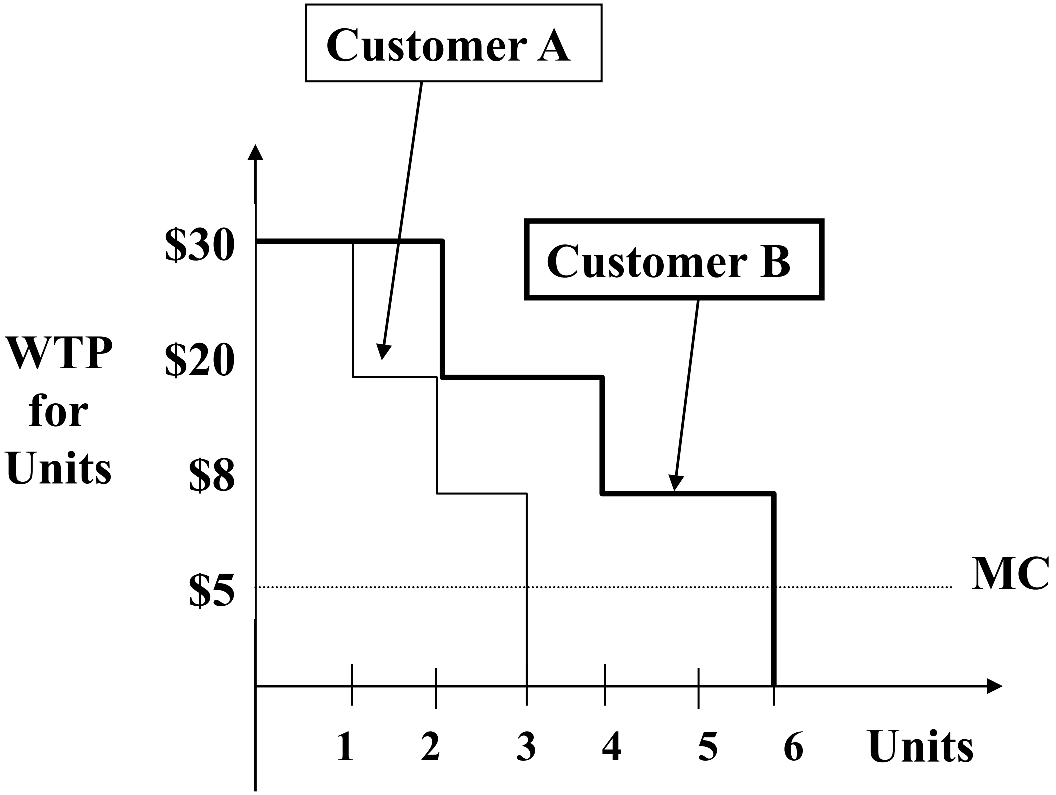
digital service; very common to see 2-part tariff due to the nature of lesser arbitrage.

- Perishability can limit such arbitrage
  - E.g., (digital) services industry can use non-linear pricing effectively because difficult/impossible to re-sell service

- Appendix: Robinson-Patman Act and **legality** of B2B volume discounts in US retail channels

Example 2:

Two customers, multiple units



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Two customers, multiple units

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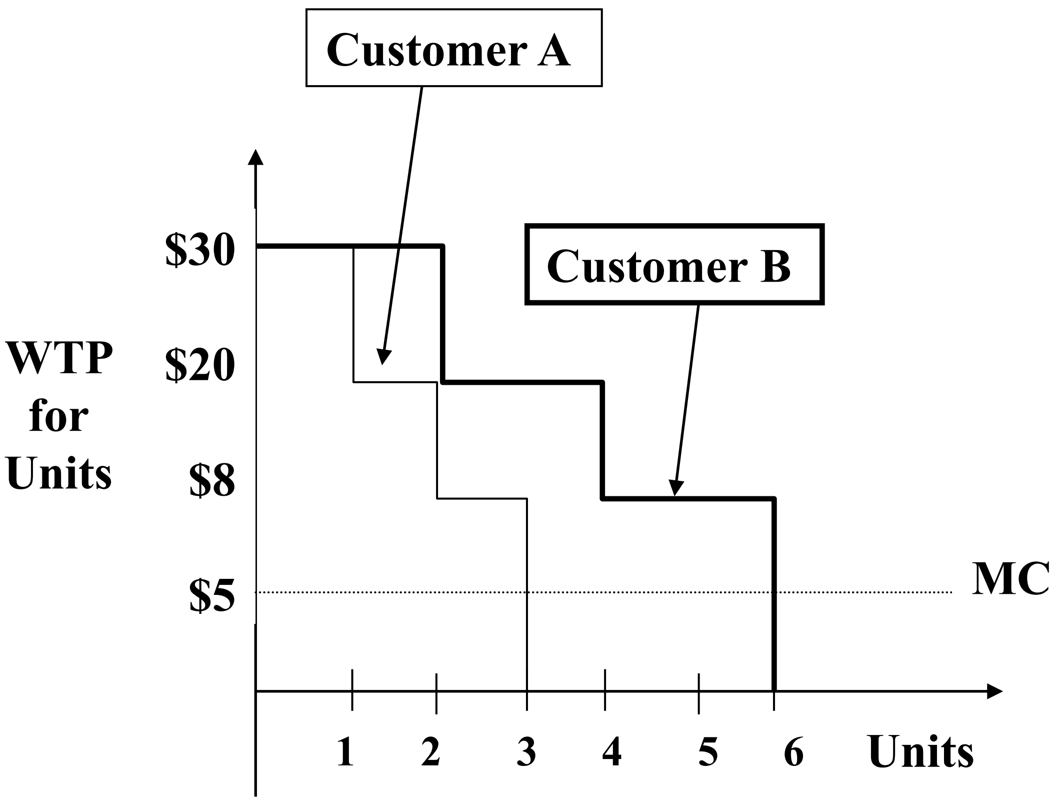

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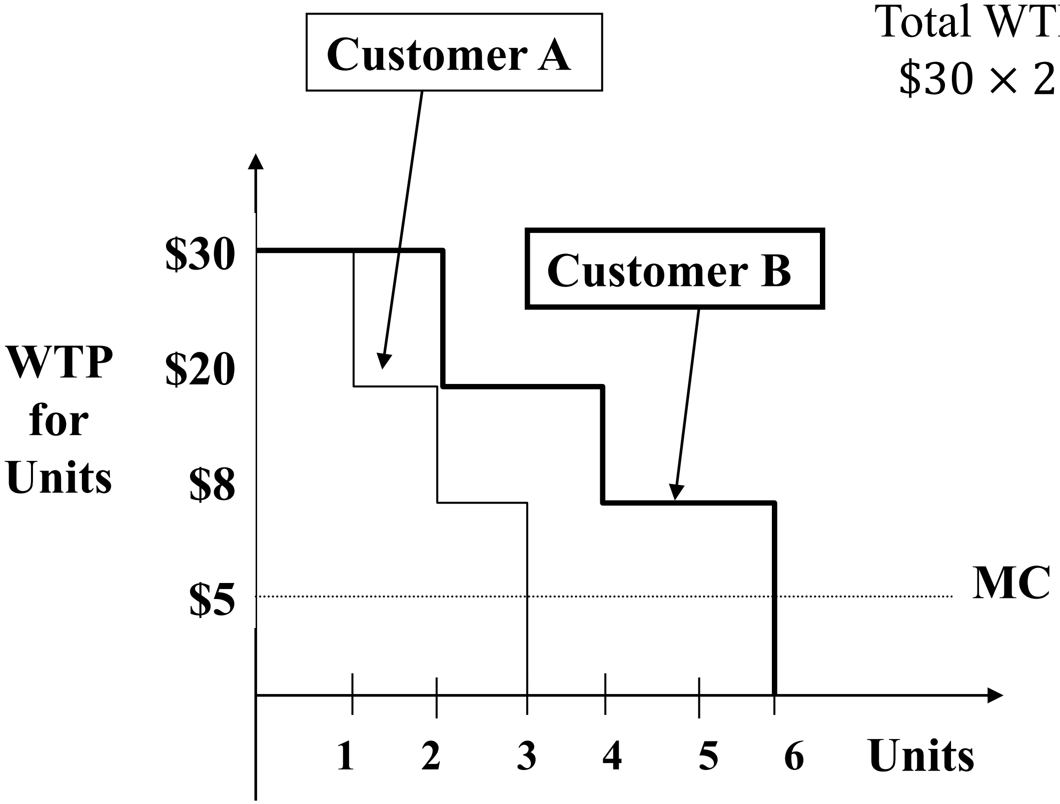

Total Customer B Value:

Total WTP for Customer B = Total Surplus

$\$30 \times 2 + \$20 \times 2 + \$8 \times 2 = \$116$

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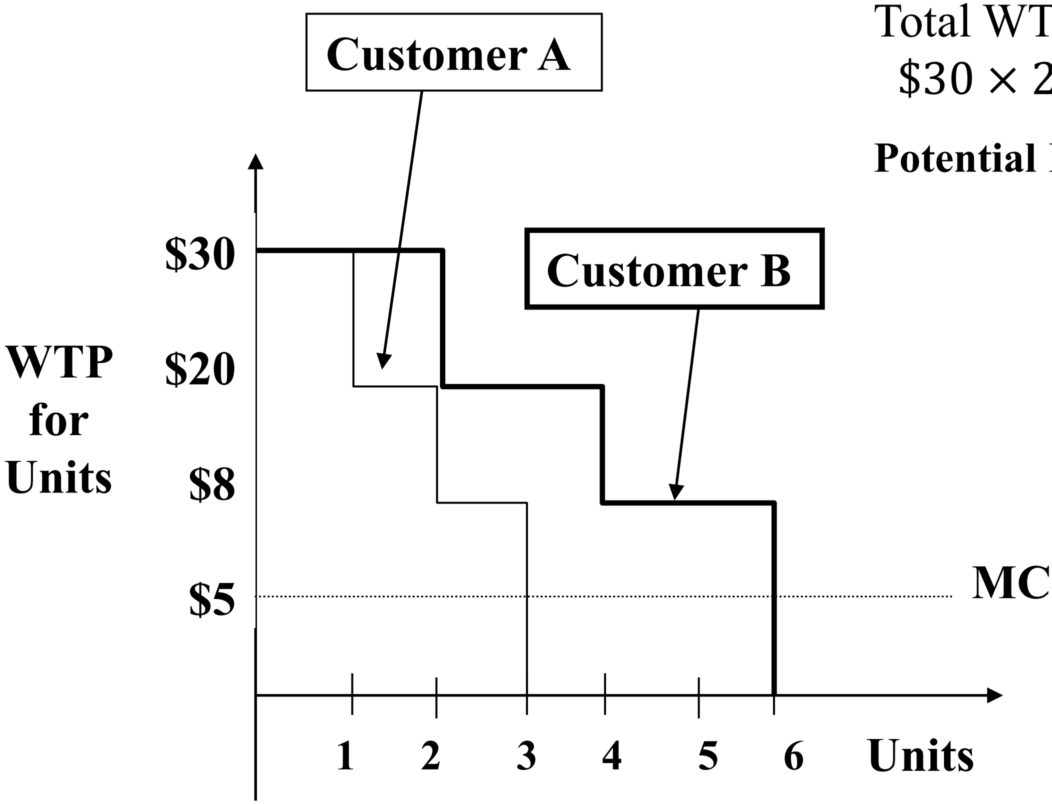

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Potential Profit in Customer B:

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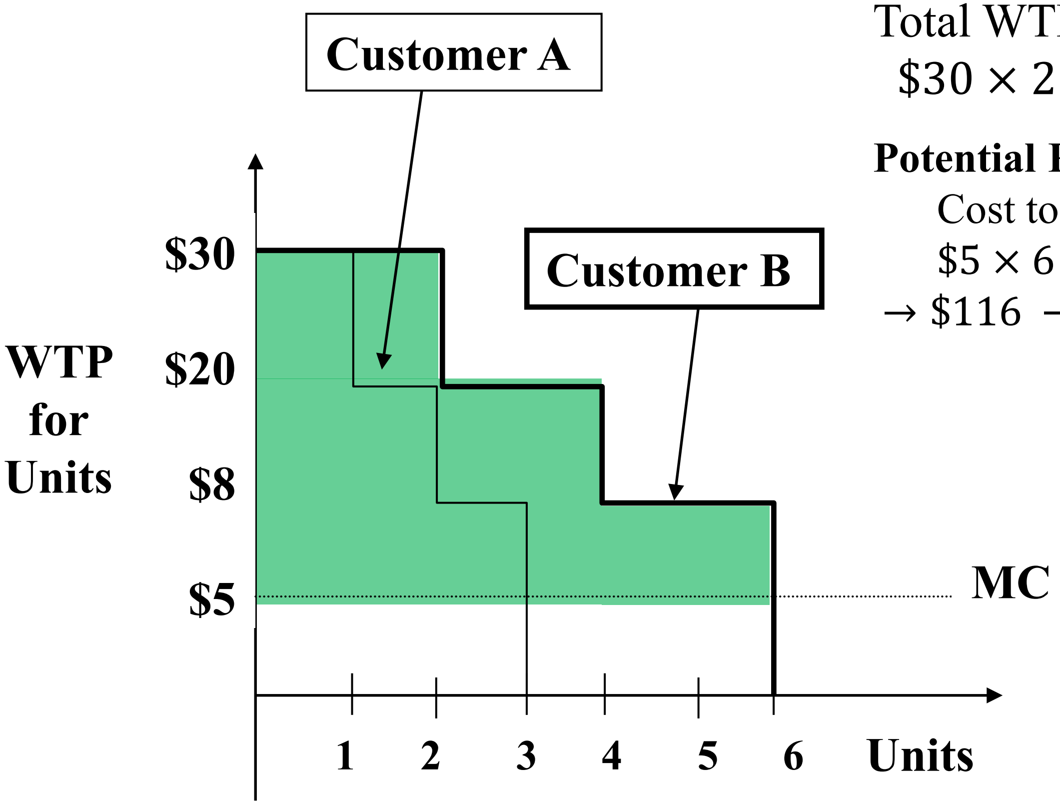

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Potential Profit in Customer B:

Cost to Produce 6 units

$\$5 \times 6 = \$\,30$

$\rightarrow \$116 - \$30 = \$86$

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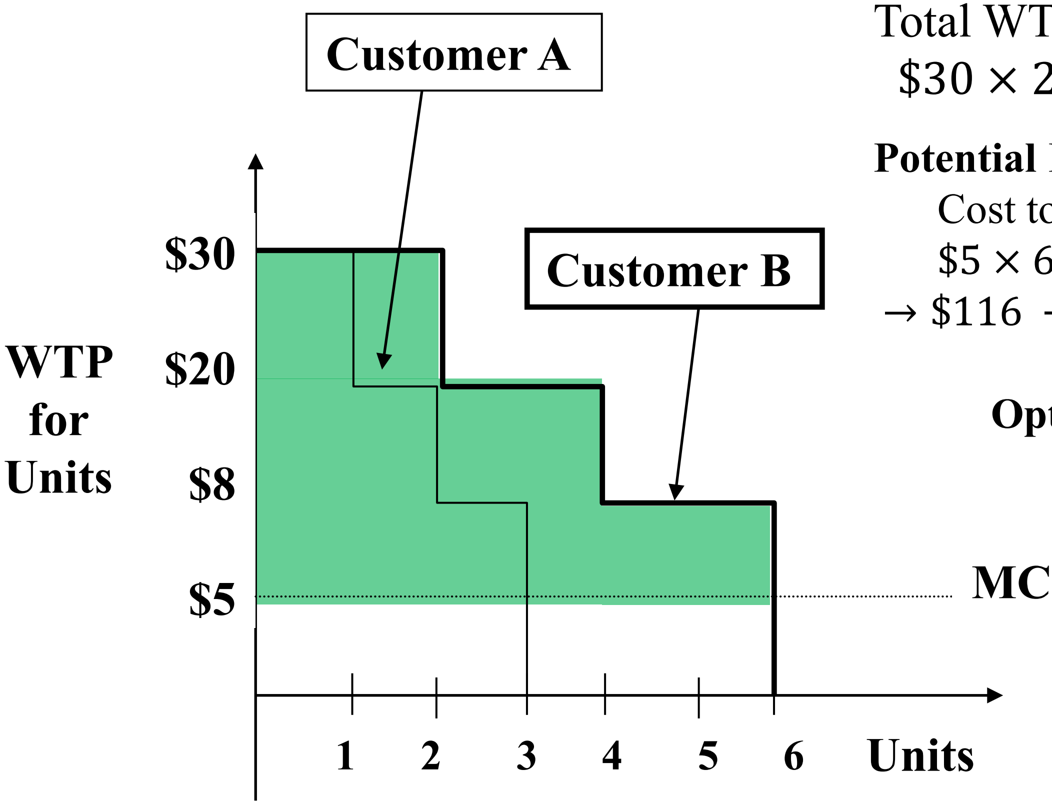

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Optimal Uniform Price (both customers):

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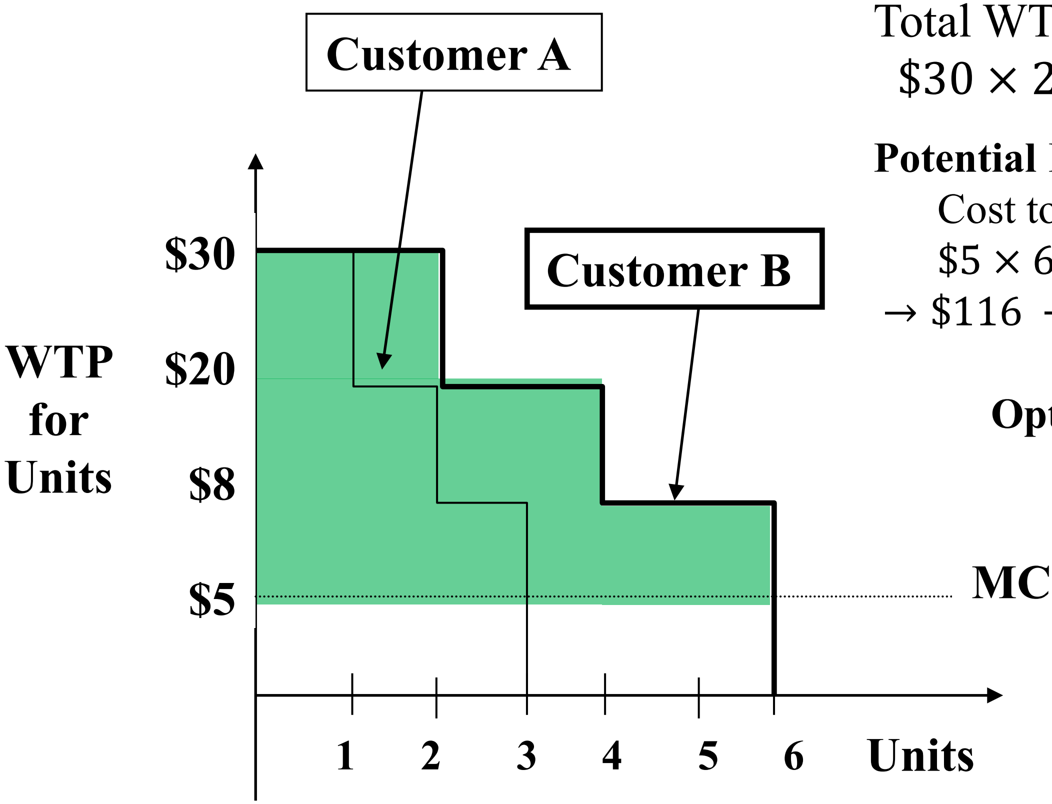

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Optimal Uniform Price (both customers):

p = $20/unit, $sell\ 6\ units$

$Profit = \$15/unit \times 6\ units = \$90$

Example 2:
Profits from both customers

- **TOTAL Potential Profit:**
  - \$43 + \$86 = \$129

    (A) (B)

- **Uniform (linear) price: p = \$20**
  - sell 6 units (2 to A, 4 to B)
  - earn \$90 in profit

⇒ leave (\$129-\$90) **\$39** potential profit on table!

Example 2:
Look at Total WTP for n units

Customer A		
Units	WTP for n^{th} unit	Cumulative WTP
1	30	30
2	20	50
3	8	58

Customer B		
Units	WTP for n^{th} unit	Cumulative WTP
1	30	30
2	30	60
3	20	80
4	20	100
5	8	108
6	8	116

My thought; if we charge a 3-pack at $58, customer B will buy just 1 of these pack. because if they buy 2 3-pack, their utility is zero, if they buy 1 pack, their utility is 80-58 = 22 (?)

JP teach; EVC of B for 6 pack = RV + DV = 58 (three pack) + 116-80 (6 pack minus utility of three pack) = $94

Another way (by Jeff), when B buys a 3 pack, he derives 22 utility. So i must give him the same 22 utility if he buys a 6 pack, which is $116-22 = $94. same answer.

Example 2:
Strategy 1

- Suppose we offer Customer A:

$$p^{3\ units} = \$58$$

- How much surplus does Customer B get from above offer?
  - For 3 units, Cumulative WTP for Customer B is \$80

  $\longrightarrow Surplus = \$80 - p^{3\ units} = \$22$

- Suppose we want Customer B to choose 6 units. What price can we set? (use EVC)
  - Use value for 6 units = \$116
  - Discount by \$22 to prevent "Trading Down"

$$p^{6\ units} = \$\ 94$$ (EVC = DV+RV = (116-80)+58)

- Total Profits: \$152 - \$45 = **\$107**

Example 2:
Can we do better?

- How did we manage *trading-down* in strategy 1?
- Is this the only lever to manage customers' incentives?
- How else can we manage customers' incentives?

Customer A		
Units	WTP for n^{th} unit	Cumulative WTP
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Customer B		
Units	WTP for n^{th} unit	Cumulative WTP
1	30	30
2	30	60
3	20	80
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5	8	108
6	8	116

Example 2:

In product line pricing, we are counting on our customers to self select into their own desired package.. Therefore we have to leave some money on the table to incentivise them to select.

Strategy 2

- Suppose we offer Customer A:

$$\boldsymbol{p^{2\ units} = \$50}$$

- How much surplus does Customer B get from the above offer?
  - For 2 units, Cumulative WTP for Customer B is \$60

  $\rightarrow Surplus = \$60 - p^{2\ units} = \$10$

- Suppose we want Customer B to choose 6 units. What price can we set?
  - Use value for 6 units = \$116
  - Discount by \$10 to prevent "Trading Down"

$$\boldsymbol{p^{6\ units} = \$106}\ \text{(EVC = DV+RV = (116-60)+50)}$$

- Profits: \$156 -\$40 = **\$116**

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Example 2:
Strategy 2 (b)

- In practice, consumers may not be able to predict their quantity needs and a menu of take-it-or-leave-it **commodity bundles** may be undesirable

- As an exercise see if you can convince yourself that the following menu of **two-part tariffs** can also work:

$$T_1(Q) = \$10 + \$20Q$$

or

$$T_2(Q) = \$76 + \$5Q$$

More common pricing structure in services (e.g., voice/data/cloud)

Two uses of NL pricing to capture value

- Our examples illustrated:

1 With 1 customer, non-linear pricing "discriminates" across each unit sold

2 With multiple customers, a menu of non-linear prices "discriminates" **between** customers (segmentation)

- Note: multiple non-linear pricing structures with same profit

Menus of non-linear pricing in practice

Example: Costco

- Costco offers a menu of **<u>2-part tariff</u>** style memberships
  - Executive Membership: $110/yr + 2% Reward
  - Gold Star Membership: $55/yr

Menus of non-linear pricing in practice
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"When others were raising their chicken prices from $4.99 to $5.99, we were willing to eat, if you will, $30 to $40 million a year in gross margin by keeping it at $4.99"

Richard Galanti, CFO, Costco

Menus of non-linear pricing in practice

Example: Costco

Strategy; is to earn money from club fees. 73% of 2024 GP is $4.8B (updated)

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  - **Sold 106 million chickens in 2021** *(investor report)*

- **Subscriber fees:** 60% of 2021 operating profit (**$6.7 Billion!**)
- **Membership:** 100 million worldwide (2022)
- **Retention rate:** 94% (2020)

“Retailer” versus “Club”?

Menus of non-linear pricing in practice

Example: Costco

Rotisserie is a best idea, its cook visually and people who shops can smell it.

- How Costco sustains $4.99 price in 2022, under inflation? (i.e., no cost pass-through yet)
  - Most retailers now charger $6-$10
- Vertical integration:

Costco's inflation-proof $4.99 rotisserie chicken, explained

Vox, July 18, 2022

  - Raises several ethical concerns on humane sourcing of food

- See appendix on Costco's low inflation cost pass-through

Flat-Rate Bias and Consumer Psychology

"New Year's Resolutions. I WILL [...] go to the gym three times a week not merely to buy sandwich."
- Bridget Jones's Diary: A Novel

All-you-can-eat
vs
Pay-as-you-go

"Monday 28 April. [...] no. of gym visits so far this year 1, cost of gym membership per year £370."
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A study of 7,500 gym members showed

- majority chose monthly fee instead of per-visit
- over-paid by over 70%

Source: DellaVigna and Malemendier (2006)

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Flat-Rate Bias & Consumer Psychology

flat rate bias – customers have a *preference* for a subscription (i.e., *all-you-can-eat*).

i.e., prefer to pay flat fee that costs more than metered service

Motivational Reasons

- Pre-commitment
- Risk aversion

Plus, the sunk cost fallacy. when i already spent irreversible money, that money is irrelevant. but we dont feel this way in reality.

risk aversion; my job is to get potential max util to keep within company budget. thus i am likely to overpay on the fixed price buffet.

Cognitive Reasons

- Framing problem
- Difficulty estimating probabilities
- Usage uncertainty/inattention

Consumers have an intrinsnic utility from the 'all you can eat' choice.

therefore, people are willing (subconsciously?) to pay a premium for buffet type services.

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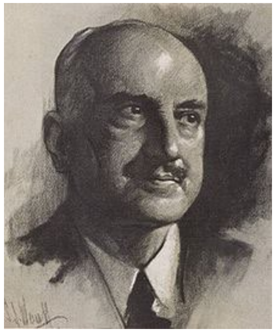
Perils of Flat-Rate Pricing

Unlimited (all-you-can-eat) Data Plans

"Those who cannot remember the past are condemned to repeat it."



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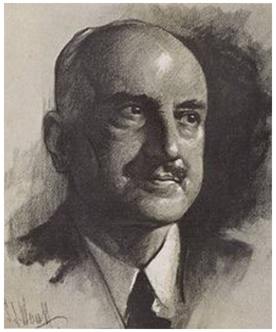
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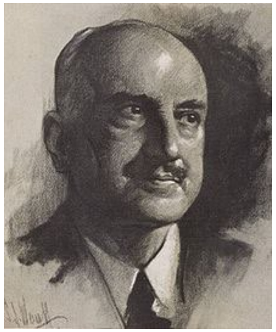

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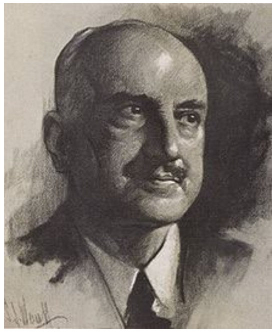
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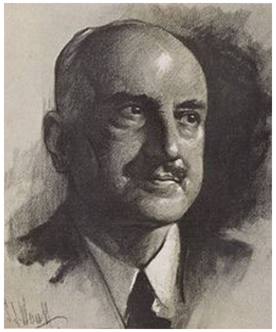

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- Verizon famously eliminated *all-you-can-eat* data in 2011!
- Competitive pricing pressure (prisoner's dilemma ?)

Proliferation of Subscriptions

Everything's becoming a subscription, and the pandemic is partly to blame

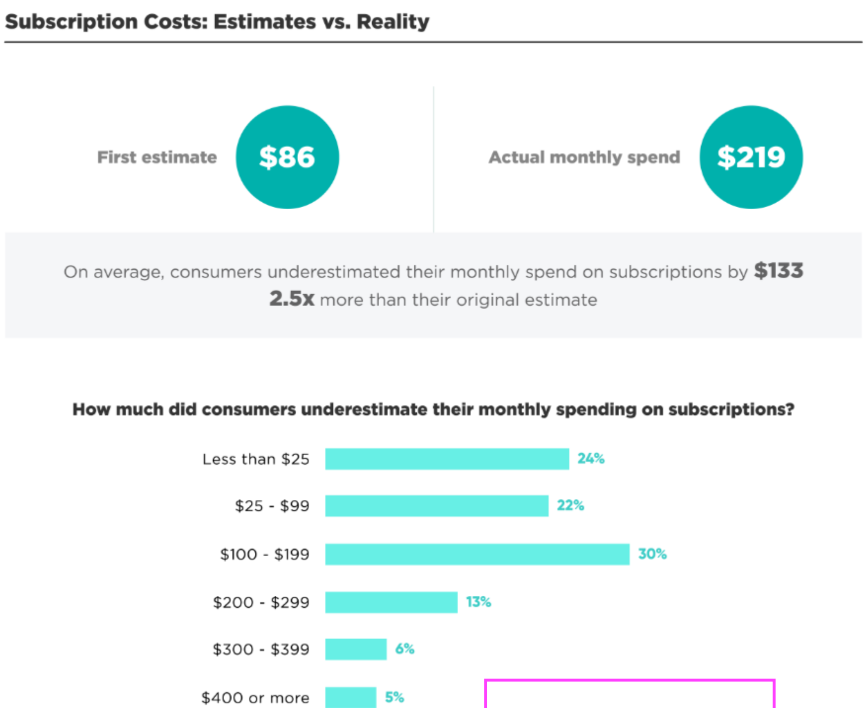
Delivery services, socks, razors, gyms, streaming services, and even restaurants and car washes. Some Americans are now signed up for 10 or more. It's bringing convenience — and a lot of monthly fees.

By Heather Long & Andrew Van Dam, *Washington Post*, June 1, 2021

"The UBS financial services firm predicts that this 'subscription economy' will grow to $1.5 trillion by 2025, more than double the $650 billion it's estimated to be worth now."

Marketing for Good?

Subscriptions & **consumer inattention**



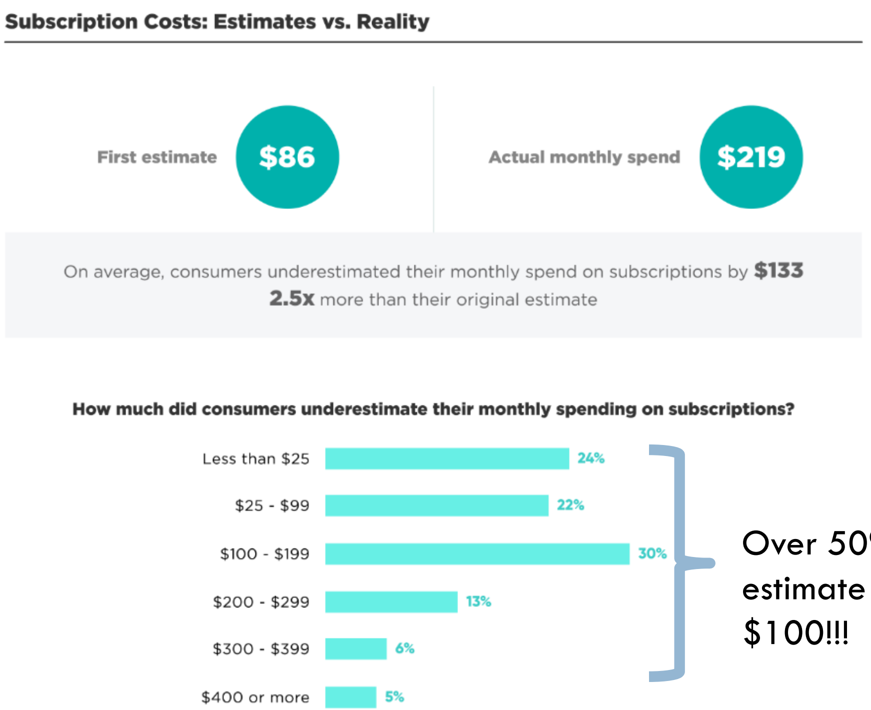
Consumers just 'forget' on how much they spend each month on subscriptions.

Source: C+R Research

The University of Chicago Booth School of Business

Marketing for Good?

Subscriptions & **consumer inattention**



Over 50% under-estimate by at least $100!!!

[Source: C+R Research]()

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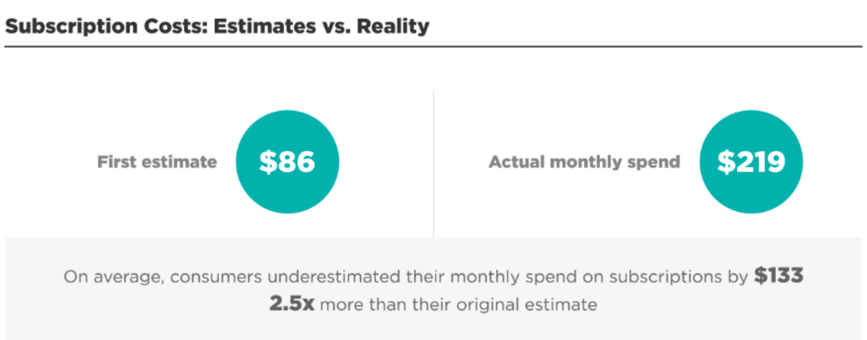

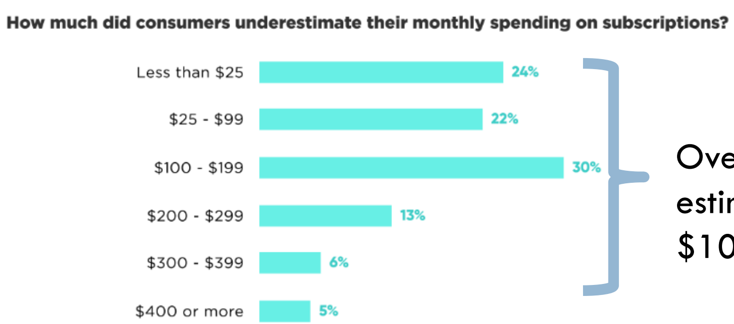

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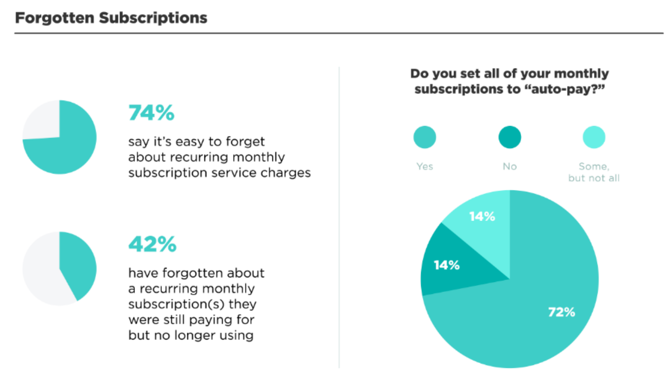

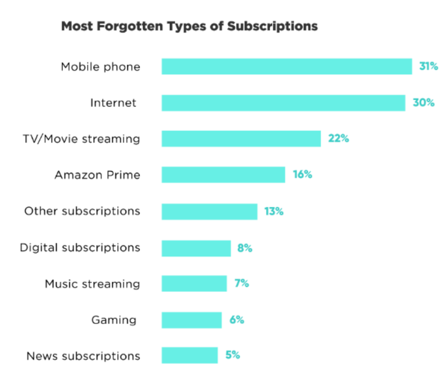

Source: C+R Research

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Marketing for Good?

Subscriptions & **consumer inattention**

How subscriptions earn excess margins?

Front end; exploit our flat rate bias
continual; prey on our consumer inattention.

- Auto-pay creates psychological switching costs…

  until it doesn't any more …

- **Consumer solutions:**

"Subscription Price Creep Is Real. Our Guide to Pushing Back."

Track down hidden recurring costs, binge strategically and consider free alternatives

Nicole Nguyen, *WSJ*, April 23, 2023

- **Regulatory solutions:**

Federal Trade Commission Proposes Rule Provision Making it Easier for Consumers to "Click to Cancel" Recurring Subscriptions and Memberships

Proposal seeks to make it as easy to cancel enrollment as it was to sign up

FTC, March 23, 2023

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Take Away

- NL pricing as price discrimination:
  - For a single customer, NL pricing captures value
  - For multiple customer segments, menu of NL pricing offers as ***product-line pricing***
- Consumer psychology as implementation guide for NL pricing
- See Appendix for agency issues with NL pricing in financial services

Marketing for good:

Economic logic for NL pricing
vs
Exploitation of inattention using complex pricing structure

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NL Pricing of Financial Services

NL Pricing in Financial Services
Example: *fee-only* versus *fee-based*

- Traditional brokerage uses 2-part tariffs (fee+commission)
- Recent surge in demand for *fee-only* service *(all-you-can-eat)*.
- Caution:
  1. *Deception* of "*fee-based*" pricing typically just 2-part tariff
  2. NL pricing affects the incentives of your asset manager!
     - Commissions incent more transactions
     - Fees tied to principal weaken incentives (i.e. less *effort*)
     - Fees tied to returns incents risk-taking

- **NL pricing special in this industry because you also need to understand the incentives induced by the pricing of your service for your advisor**

- Sources:

"Much Ado About Adviser Compensation," Daisy Maxey, *WSJ*, Nov 10, 2013.

"How to Pay Your Financial Advisor," Daisy Maxey, *WSJ*, Dec 12, 2011

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Legality of NL Pricing in US

Non-Linear Pricing and B2B Laws
The Robinson-Patman Act in the U.S.

- *Act requires Mfrs to sell products to all retailers at the same price – unless there is something significantly different (e.g. large-size packaging at Costco)*
  - e.g. 32-packs of Coke, 3-packs of printer cartridges
- Historically - protect small independent retailers (FTC vs Morton Salt Co.)
- FTC now worried act shields less efficient competition
  - **Why it's all in the packaging: 1936 antitrust ruling comes under fresh criticism**
    John Schmeltzer. ***Knight Ridder Tribune Business News. Jun 7, 2007.***
  - **Third Circuit Reverses Price Discrimination Judgment in *Feesers v. Michael Foods and Sodexho*** **.** 08-January-2010
- If abolished, big-box chains will surely command better volume discounts

Non-Linear Pricing and B2B Laws
The Robinson-Patman Act in the U.S.

- Precedent may change under Lina Khan's oversight of FTC

Pepsi, Coke soda pricing targeted in new federal probe

By Josh Sisco, *Politico*, Jan. 9, 2023

- "Beverage giants Coca-Cola and PepsiCo are under preliminary investigation at the Federal Trade Commission over potential price discrimination in the soft drink market as the agency looks to revive a decades-old, but largely dormant law banning the practice, according to four people with knowledge of the probe. …For at least the last month the FTC has reached out to large retailers, including Walmart, seeking data and other information on how they purchase and price soft drinks, two of the people said. Walmart is not currently a target in the investigation, one of the people said."

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Inflation and Cost Pass-Through on Costco Chicken

Retail Loss-Leader Pricing

Inflation is raising prices on almost everything, except rotisserie chicken

By Rachel Treisman, *NPR*, June 7 2022

- Chicken prices up 16.4% in 2022, & predicted to increase 15-18%.
- But BJ's, Costco and Sam's still charge under $5
- Chains like Publix and Giant Eagle charge $7

"Rotisserie chickens in particular are a marketable combination of visible and Affordable…They're giving a cheapness aura to the store, which is helpful for consumers when they're concerned about their spending power.

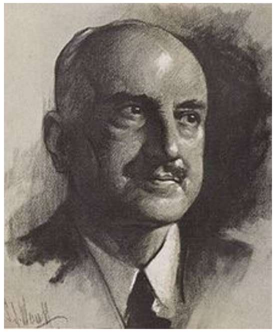
Flat-Rate Bias

Mobile Voice and Data Services

Perils of Flat-Rate Pricing

Unlimited (all-you-can-eat) Data Plans

Those who cannot remember the past are condemned to repeat it.



George Santayana

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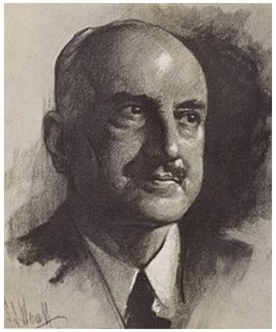
George Santayana

Why every US carrier has a new unlimited plan

by Chaim Gartenberg@cgartenberg, *The Verge*, 2-17-2017

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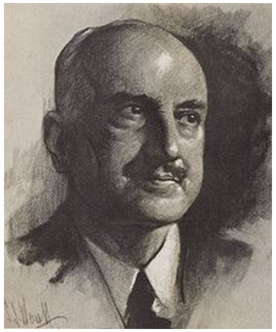

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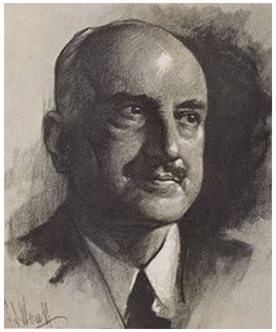

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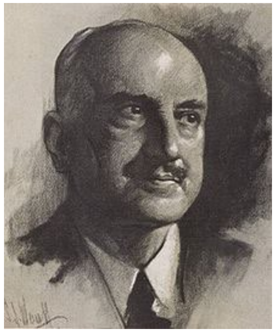

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- Competitive pricing pressure (prisoner's dilemma ?)